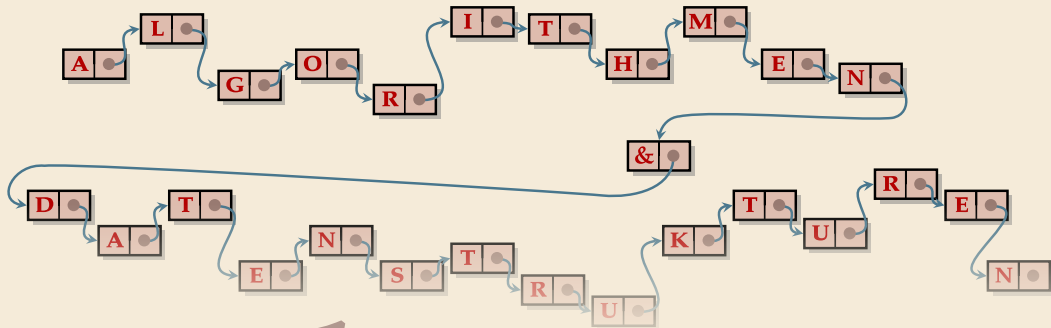




4

Case Study: Union-Find

Algorithmen & Datenstrukturen · Sommersemester 2026

Prof. Dr. Sebastian Wild

4 Case Study: Union-Find

- 4.1 Dynamic Connectivity
- 4.2 Quick Find
- 4.3 Ist Quick Find gut genug?
- 4.4 Quick Union
- 4.5 Verbesserungen
- 4.6 Anwendungen & Perkolationstheorie

Ausblick

Bevor wir noch etwas systematischer auf Algorithmenanalyse und Maschinenmodelle schauen, erst einmal ein konkretes Anwendungsbeispiel!

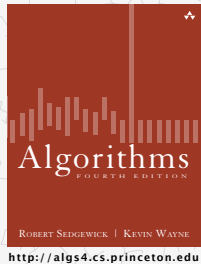
Ziele

- ▶ Schritte des Algorithmenentwurfs illustrieren
 1. Problem modellieren
 2. Ersten/einfachen Algorithmus für das Problem finden
 3. Schon schnell genug? Passt in den Speicher?
 4. Falls nicht, Engpässe finden nicht immer einfach; manchmal sogar unmöglich ... dazu später mehr!
 5. Verbesserten Algorithmus finden, ggf. iterieren ↙
- ▶ Potential guter algorithmischer Ideen zeigen!
- ▶ Erstes Werkzeug für Ihren Werkzeugkasten

Heutiges Material von Sedgewick & Wayne

Algorithms

ROBERT SEDGEWICK | KEVIN WAYNE



1.5 UNION-FIND

- *dynamic connectivity*
- *quick find*
- *quick union*
- *improvements*
- *applications*

4.1 Dynamic Connectivity

Dynamic connectivity problem

Given a set of N objects, support two operation:

- Connect two objects.
- Is there a path connecting the two objects?

connect 4 and 3

connect 3 and 8

connect 6 and 5

connect 9 and 4

connect 2 and 1

are 0 and 7 connected? ✗

are 8 and 9 connected? ✓

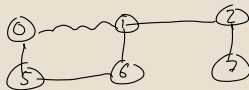
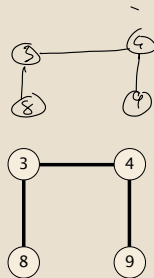
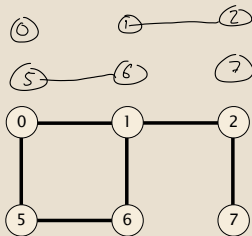
connect 5 and 0

connect 7 and 2

connect 6 and 1

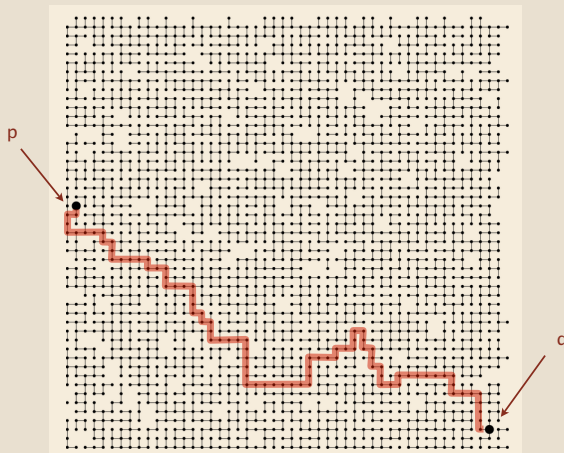
connect 1 and 0

are 0 and 7 connected? ✓



A larger connectivity example

Q. Is there a path connecting p and q ?



A. Yes.

Modeling the objects

Applications involve manipulating objects of all types.

- Pixels in a digital photo.
- Computers in a network.
- Friends in a social network.
- Transistors in a computer chip.
- Elements in a mathematical set.
- Variable names in a Fortran program.
- Metallic sites in a composite system.

When programming, convenient to name objects 0 to N - 1.

- Use integers as array index.
- Suppress details not relevant to union-find.

 can use symbol table to translate from site names to integers: stay tuned (Chapter 3)